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RESEARCH ARTICLE

Tension-Based Reconfigurable Multi-Agent Formation for Aerial Load Transportation

VINÍCIUS PACHECO BACHETI¹, DANIEL K. D. VILLA¹,
ROGELIO LOZANO², (Senior Member, IEEE), MÁRIO SARCINELLI-FILHO¹,
AND PEDRO CASTILLO², (Member, IEEE)

¹Department of Electrical Engineering, Universidade Federal do Espírito Santo, Vitória 29075-910, Brazil

²Heudiasyc Laboratory, CNRS UMR 7253, Université de Technologie de Compiègne, 60200 Compiègne, France

Corresponding author: Mário Sarcinelli-Filho (mario.sarcinelli@ufes.br)

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ABSTRACT This paper proposes a trajectory tracking controller for a cable-suspended load transported by n uncrewed aerial vehicles (UAVs) based on Euler angle states. This is accomplished using the resultant tension on the load as a virtual input and then distributing it to each vehicle, for which low-level controllers generate the correct thrust and torque values. Furthermore, a leader-follower-based control is proposed to safely change the number of agents carrying the load. Numerical simulations are presented to validate the controllers. Finally, experiments demonstrate that the proposed method effectively enables the load to perform as expected and allows for the configuration of the formation to be changed without significantly impacting the load navigation.

INDEX TERMS Autonomous aerial vehicles, cooperative systems, load modeling, mobile robots, multi-robot systems.

I. INTRODUCTION

As technology advances and becomes more affordable, the use of robots in different applications increases. Initially developed for military applications, drones are now utilized in various civil applications, including disaster management [1], crop monitoring [2], marine fauna detection [3], habitat destruction assessment [4], and many others. Among these applications, one that has received increasing attention is the transportation of loads by uncrewed aerial vehicles (UAVs). As recent surveys [5], [6] show, multirotor vehicles have several advantages concerning this kind of application, such as maneuverability, the ability to hover in place and operate in a confined environment, and ease of use. However, they also come with disadvantages, namely short flight time and small payload capacity [7].

Usually, for multirotors, load transportation is achieved using grippers rigidly attached to the body of the multirotor [8], [9] or cables [10], [11]. While the former can increase the vehicle's inertia, making it more sluggish for fast attitude response, the latter increases the degree of underactuation, making its control and motion planning more difficult [12]. The use of cables, however, has the significant advantage of enabling multiple multirotors to carry a single load cooperatively.

Teams of drones have been used for a long time to compensate for the drawbacks of a single aircraft, such as to cover a wider area considering simultaneous different viewpoints or to share sensors to reduce the collective system payload [13]. Thus, it is natural to use multiple drones to carry loads. While this makes a difficult problem even more complex, it comes with the benefits of increasing the possible weight of the payload, reducing oscillations, as well as possibly enabling the positioning and orienting of the load, depending on the number of drones and anchor points [14].

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Several recent works have tackled this kind of system. In [15], a method is proposed to model and optimize cable tensions in a multi-UAV system transporting a payload. It employs a mass-spring-damper approach to model the dynamics of flexible cables connecting the UAVs to the payload, taking into account elasticity, damping, and slack conditions. A Nonlinear Least Squares (NLS) regression algorithm, based on the Gauss-Newton method, is employed to fine-tune the parameters of the tension model. The methodology also incorporates control strategies, such as PID and ISMC, to stabilize the system during these events.

In [16], the authors present a control approach for a multi-UAV system, named the Fly-Crane setup, designed to manage a cable-suspended load. The methodology prioritizes robust full-pose control (position and orientation) in the presence of uncertainties in the model and external disturbances. It simplifies the process by focusing on kinematic parameters, which are easier to measure accurately and precisely. The system is divided into inner and outer control loops: the inner loop uses decentralized proportional velocity controllers for each UAV, while the outer loop generates reference velocities for the UAVs to maintain the load's configuration.

Both [17] and [18] leverage fuzzy logic approaches to enhance the performance of uncrewed aerial systems in load transportation. The former does it by proposing a system, termed ELD-FFSMC, which integrates an equilibrium load distributor with a fuzzy sliding-mode controller to enhance robustness and precision in trajectory tracking. This approach addresses uncertainties and external disturbances, ensuring finite-time convergence of errors through a PID sliding surface, as validated by the Lyapunov method. The system's distributed nature eliminates the need for information sharing among agents, relying solely on payload position for guidance. The latter considers the payload as a disturbance with unmodeled dynamics affecting the aircraft and introduces a control strategy that utilizes a fuzzy approximation technique to address such unknown dynamics. Moreover, it enhances this controller with a novel consensus-inspired approach that enables two UAVs to cooperatively perform the load transportation task in a coordinated manner.

Work is also being done with differently shaped loads, such as in [19], [20], [21], where two drones carry a payload in the shape of a rod. Another recent example is [22], which presents a cooperative control method for two UAVs transporting a bar-shaped payload connected via cables without relying on inter-UAV communication. The leader UAV controls the payload trajectory using sensor data from an inertial measurement unit (IMU) attached to the payload, while the follower adjusts its motion to maintain stability. The system models the payload as nonholonomic to restrict certain movements, ensuring that the follower's motion does not interfere with the leader's trajectory tracking. Unscented Kalman Filters estimate forces and positions, which removes the need for physical force sensors.

In [23], a multi-UAV system transporting a payload with flexible cables is considered, addressing how to handle the failure of one UAV during flight. The proposed method involves modeling the system dynamics without simplifications and creating a recovery strategy using PID, Integrative Sliding Mode Control (ISMC), and Consensus-Based control. The system's dynamics are modeled using equations for forces, torques, and cable tensions, employing a mass-spring-damper approach. The control design ensures that the remaining UAVs redistribute the load and stabilize the payload by directly acting on cable tensions.

One of the challenges that a team of drones cooperatively carrying a single load faces is the localization of the load and of the agents, both in relation to the world and to each other, since the cables attached to each element of the formation result in heavily coupled dynamics. Some works [24], [25], [26] use cameras embedded on the drones pointed downwards in order to localize the load in relation to them, while other works [27], [28] use different approaches to solve the localization problem, such as barycentric coordinates or bipartite tracking consensus.

In this work, we have chosen to tackle the problem of controlling a load attached to multiple drones via cables. This will be accomplished by utilizing the resultant tension on the load as a virtual control signal, and then distributing this tension to each UAV, where low-level controllers will generate the correct thrust and torque values. We also discuss modifying the number of agents during mission accomplishment using a leader-follower-based control. Such a modification may be necessary to accommodate handling agent failures or reduce the robot cluster's size to comply with possible physical environmental constraints, for instance.

The failure of a robot could cause it to become part of the payload, making the recovery attempt more difficult. Nevertheless, if it is possible to detach such a UAV from the system, as [29] hypothesizes, then the controller presented in this work could be a powerful tool to ensure complete mission accomplishment. However, this work will focus more on reconfiguring the formation by adjusting the number of agents carrying the load to accommodate new environmental constraints, such as navigating a narrow corridor.

Therefore, the main contributions of this work are as follows: (i) a closed-loop trajectory tracking controller for a cable-suspended load system with n agents, designed based on Euler angle states and closed on the load itself; and (ii) a leader-follower control strategy capable of safely managing dynamic changes in the number of agents involved in load transportation. Notably, the proposed controller can restore the original formation after a change in agent configuration, as demonstrated in the experimental results. Additionally, numerical simulations with $n > 4$ agents are provided further to validate the effectiveness and scalability of the proposed controllers. Finally, to the authors' knowledge, this is the first work to address dynamic agent reconfiguration for load transportation in a structure of this kind. Three

distinct experiments are presented to illustrate and validate the proposed approach.

This work continues as follows: in Section II, the problem being addressed is characterized, along with the translational and rotational system models. Section III describes the controllers responsible for tracking the load trajectory, while Section IV discusses how the number of agents is modified. The results are presented in Sections V and VI, where the former discusses the simulations conducted and the latter displays the experimental results obtained. Finally, Section VII synthesizes the findings and advances some future work.

II. PROBLEM DEFINITION

Consider a system of n robots attached to a load via cables, subjected to the following assumptions:

- The load is considered to be a point mass.
- The drones are attached to the load via a single cable attached to their center of mass.
- The cables are considered rigid and subjected to a positive tension force.
- The cables are massless.

Regarding the configuration variables of this system, $\mathbf{x}_0 = [x_0 \ y_0 \ z_0]^T \in \mathbb{R}^3$ is the load position relative to the inertial system of coordinates, $\mathbf{x}_i = [x_i \ y_i \ z_i]^T \in \mathbb{R}^3$, with $i = 1, 2, \dots, n$, is each drone's position, $l_i \in \mathbb{R}$ is the length of the cable connecting drone i to the point mass and $\alpha_i \in \mathbb{S}^2$ is a unitary vector representing the direction of the tension force in l_i . In addition, each drone has an orientation defined by $\eta_i = [\phi_i \ \theta_i \ \psi_i]^T \in \mathbb{R}^3$, where each variable stands for the roll, pitch and yaw of the vehicle, respectively. A graphical representation of the system for $n = 3$ is illustrated in Fig. 1.

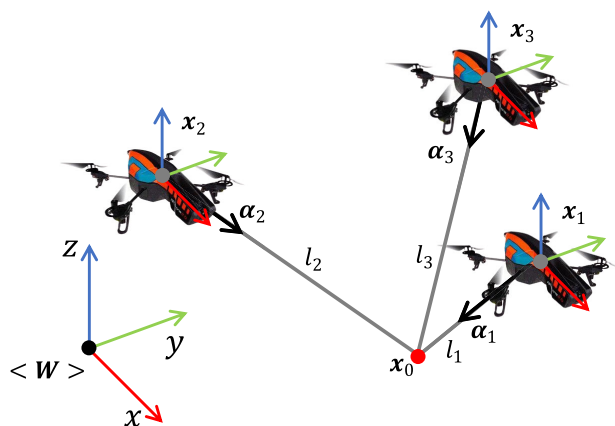


FIGURE 1. Graphical representation of the system agents and the configuration variables for $n = 3$.

The dynamic behavior of the load is represented by

$$m_0 \ddot{\mathbf{x}}_0 = -m_0 g \mathbf{e}_3 - \sum_{i=1}^n T_i \alpha_i, \quad (1)$$

where m_0 is the mass of the load, g is the gravitational acceleration, $\mathbf{e}_3 = [0 \ 0 \ 1]^T$ is a unitary vector pointing in the z direction, and $T_i \in \mathbb{R}$ is the tension in cable i .

From Fig. 1, and the fact that the cables are rigid with constant length l_i , notice that the load is subjected to the restriction

$$\mathbf{x}_i = \mathbf{x}_0 - l_i \alpha_i, \quad (2)$$

concerning each UAV. As for the dynamic behavior of the drones, due to the premise that the cables are attached to their center of mass and to the decoupled nature of the dynamics of $\mathbf{x}_i$ and η_i coordinates [30], it is possible to claim without loss of generality that the tension force does not impact the angular dynamics directly, and thus, based on [31], it is possible to write the translational dynamic equation of the drones as

$$m_i \ddot{\mathbf{x}}_i = f_i \mathbf{R}(\eta_i) \mathbf{e}_3 - m_i g \mathbf{e}_3 + T_i \alpha_i, \quad (3)$$

where m_i is the mass of the i^{th} vehicle, $f_i \in \mathbb{R}$ is the thrust it generates, and $\mathbf{R}(\eta_i) \in SO(3)$ is its forward kinematic matrix, given by

$$\mathbf{R}(\eta_i) = \begin{bmatrix} c_{\theta_i} c_{\psi_i} & s_{\phi_i} s_{\theta_i} c_{\psi_i} - c_{\phi_i} s_{\psi_i} & s_{\phi_i} s_{\psi_i} + c_{\phi_i} s_{\theta_i} c_{\psi_i} \\ c_{\theta_i} s_{\psi_i} & c_{\phi_i} c_{\psi_i} + s_{\phi_i} s_{\theta_i} s_{\psi_i} & c_{\phi_i} s_{\theta_i} s_{\psi_i} - s_{\phi_i} c_{\psi_i} \\ -s_{\theta_i} & s_{\phi_i} c_{\theta_i} & c_{\phi_i} c_{\theta_i} \end{bmatrix}, \quad (4)$$

where c_x and s_x correspond to $\cos x$ and $\sin x$, respectively.

Regarding angular dynamics, let $\Omega_i \in \mathbb{R}^3$ represent the angular velocities of drone i in its reference system, which relates to the angular velocity in the inertial system η_i through the expression $\Omega_i = \mathbf{W}_{\eta_i} \dot{\eta}_i$, where

$$\mathbf{W}_{\eta_i} = \begin{bmatrix} 1 & 0 & -s_{\theta_i} \\ 0 & c_{\phi_i} & s_{\phi_i} c_{\theta_i} \\ 0 & -s_{\phi_i} & c_{\phi_i} c_{\theta_i} \end{bmatrix}. \quad (5)$$

In our analysis, the Euler angle coordinates convention is the following: the x -axis points in the forward direction, and the z -axis points up, obtaining $[\phi_i \ \theta_i \ \psi_i]^T$.

With these definitions, the rotational dynamics of the drones can be written as

$$\ddot{\eta}_i = \tilde{\tau}_i, \quad (6)$$

where $\tilde{\tau}_i = [\tilde{\tau}_\phi \ \tilde{\tau}_\theta \ \tilde{\tau}_\psi]^T \in \mathbb{R}^3$ are the rolling, pitching and yawing moments, respectively, which are related to the input torques τ_i by

$$\tilde{\tau}_i = \mathbf{J}^{-1}(\tau_i - \mathbf{C}(\eta, \dot{\eta})\dot{\eta}), \quad (7)$$

where $\mathbf{J}$ is the inertia matrix of the drone expressed in the generalized coordinates η , and $\mathbf{C}(\eta, \dot{\eta})$ is the Coriolis term, which is

$$\mathbf{J} = \mathbf{W}_{\eta}^T \mathbf{I} \mathbf{W}_{\eta}, \quad (8)$$

and

$$\mathbf{C}(\eta, \dot{\eta}) = \dot{\mathbf{J}} - \frac{1}{2} \frac{\partial}{\partial \eta} (\dot{\eta}^T \mathbf{J}). \quad (9)$$

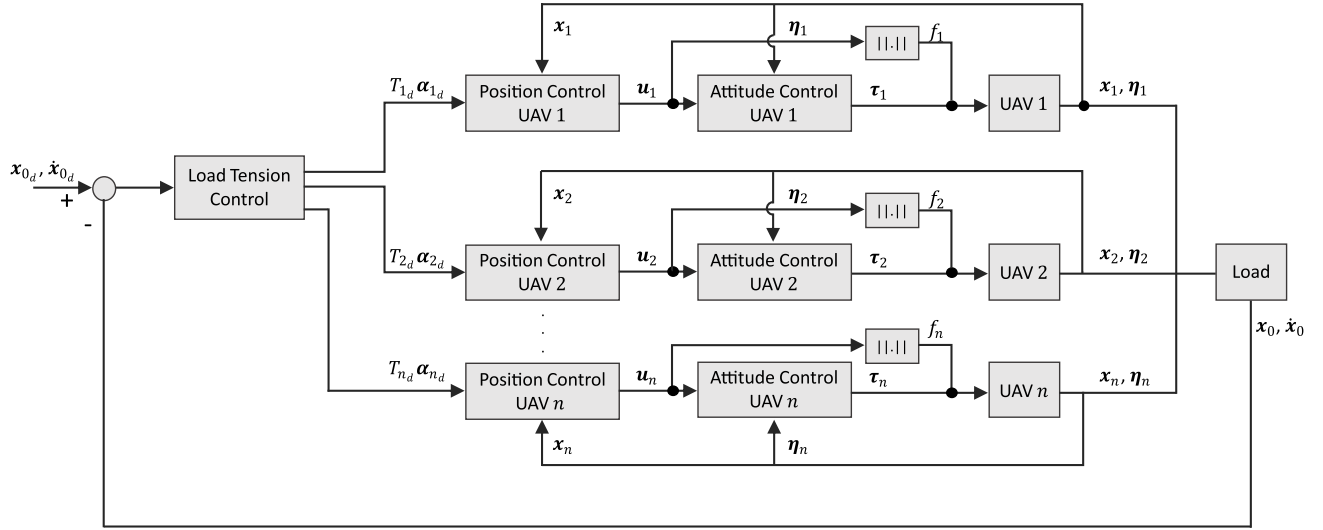


FIGURE 2. Control diagram of the n -multi-agent carrying a load.

Finally, this work considers the trajectory tracking by the load, which means that the controller should be able to reduce position and velocity errors when the load follows a moving target. The following section discusses how this is accomplished.

III. LOAD CONTROLLER

This section provides a detailed description of the controller developed for load trajectory tracking, ensuring the stability of the entire system. The Point of Interest (POI) for control is always the load position, and all trajectories are defined in its space.

Considering that the real input of the system is the thrust f_i of each drone and their torques $\tau_i = [\tau_{\phi_i} \ \tau_{\theta_i} \ \tau_{\psi_i}]^T$, the main idea of this work is to virtually control the load through considering the resultant tension in the cables as a virtual input. Then, this resultant tension is distributed among all the available UAVs, which, to generate the required tension on the cable, must assume a particular position in space, given by the desired tension vector and the length of the cable l_i , and a combination of a desired thrust and attitude. It is assumed that the desired attitude is reached quickly enough so that the UAVs converge to their desired position, which, in turn, generates the desired tensions in the cables, ultimately leading to control of the load. Figure 2 shows an overview of the control system.

The following subsections describe each of the individual steps of the controller, beginning with the load control and ending with the attitude controller based on Euler angles.

A. LOAD TENSION CONTROL

Consider the control law

$$\mathbf{u}_0 = m_0(g\mathbf{e}_3 + \mathbf{v}_0), \quad (10)$$

based on feedback linearization, where $\mathbf{v}_0$ is defined as

$$\mathbf{v}_0 = \ddot{\mathbf{x}}_{0_d} + \mathbf{K}_{p0}\tilde{\mathbf{x}}_0 + \mathbf{K}_{d0}\dot{\tilde{\mathbf{x}}}_0, \quad (11)$$

where $\ddot{\mathbf{x}}_{0_d}$ is the desired acceleration of the trajectory, $\mathbf{K}_{p0}$, $\mathbf{K}_{d0} > 0$ are diagonal gain matrices and $\tilde{\mathbf{x}}_0$ is the tracking error of the load, given by

$$\tilde{\mathbf{x}}_0 = \mathbf{x}_{0_d} - \mathbf{x}_0, \quad (12)$$

while $\dot{\tilde{\mathbf{x}}}_0$ is its derivative, given by

$$\dot{\tilde{\mathbf{x}}}_0 = \dot{\mathbf{x}}_{0_d} - \dot{\mathbf{x}}_0. \quad (13)$$

Finally, $\mathbf{x}_{0_d}$ and $\dot{\mathbf{x}}_{0_d}$ are the desired positions and velocities of the load, respectively.

Notice from (1) that the resultant tension on the load cannot be directly generated. Therefore, it can be considered as a virtual control input resulting from the tension forces of the cables, defined as the sum of all the tension forces on the cables. Hence, from (1) we can propose the control law

$$\mathbf{u}_0 = -\sum_{i=1}^n T_{i_d}\alpha_{i_d}, \quad (14)$$

where the minus sign comes from the fact that the tension forces were considered positive at the other end of the cable, as shown in Fig 1, and $T_{i_d}\alpha_{i_d}$ are the desired tension vectors generated by each drone.

Since the computation of the desired cable forces for each of the agents $T_{i_d}\alpha_{i_d}$ depends heavily on the number of agents and their spatial configuration, as well as possible additional constraints, more detailed explanations are provided in the Results section, where each specific configuration is discussed individually.

Now, define the tension error as the difference between the actual tension in the cable and the desired tension. Thus,

$$\mathbf{e}_i^T = T_i\alpha_i - T_{i_d}\alpha_{i_d}. \quad (15)$$

Then, by introducing (10), (14) and (15) into (1), the closed-loop equation takes the form

$$m_0 \ddot{\mathbf{x}}_0 = m_0 \mathbf{v}_0 - \sum_{i=1}^n \mathbf{e}_i^T. \quad (16)$$

However, if the assumption is made that the initial conditions of the system are well posed, such that the UAVs are near their ultimately desired positions relative to the load, and also that the desired positions and velocities of the load are described by smooth functions, with relatively small accelerations, it can be shown that the tension error remains bounded and slowly-varying, i.e., $\|\mathbf{e}_i^T\| \leq \delta$ and $\dot{\mathbf{e}}_i^T \approx \mathbf{0}$, where $\delta > 0$.

Observe from (16) that if $\mathbf{e}_i^T \rightarrow 0$, then

$$\ddot{\mathbf{x}}_0 = \mathbf{v}_0. \quad (17)$$

Therefore, using (11), such an equation can be rewritten as

$$\ddot{\mathbf{x}}_0 + \mathbf{K}_{d0} \dot{\mathbf{x}}_0 + \mathbf{K}_{p0} \mathbf{x}_0 = \mathbf{0}, \quad (18)$$

which is a linear form that guarantees that the velocity error $\dot{\mathbf{x}}_0$ and the position error $\mathbf{x}_0$ converge asymptotically to zero when time goes to infinity.

Thus, as (18) suggests, if the UAVs can generate, through (14), the correct tension vector in the load, given by (10), and $\mathbf{e}_i^T \rightarrow 0$, then the load will accomplish the trajectory tracking task. The following section discusses how this task is accomplished.

B. GENERATING THE DESIRED TENSION FORCE

In this subsection, the controller responsible for generating the required tension force for each drone ($\mathbf{e}_i^T \rightarrow 0$) is proposed.

To do so, consider the control law

$$\mathbf{u}_i = m_i(g\mathbf{e}_3 + \mathbf{v}_i) - T_{id} \boldsymbol{\alpha}_{id}. \quad (19)$$

Observe that $\mathbf{u}_i$ is the tridimensional force vector that the drone generates through its thrust and attitude, and so it can be written as

$$\mathbf{u}_i = f_{id} \mathbf{R}(\boldsymbol{\eta}_{id}) \mathbf{e}_3, \quad (20)$$

where $f_{id} = \|\mathbf{u}_i\|$ is the desired thrust force of each drone and $\boldsymbol{\eta}_{id}$ is its desired attitude.

By defining the UAV force vector error as the difference between the actual force generated by the combination of thrust and attitude and the desired force, it is possible to write

$$\mathbf{e}_i^u = f_i \mathbf{R}(\boldsymbol{\eta}_i) \mathbf{e}_3 - f_{id} \mathbf{R}(\boldsymbol{\eta}_{id}) \mathbf{e}_3. \quad (21)$$

This way, by combining (21) with the model given in (3), the closed-loop equation is given by

$$m_i \ddot{\mathbf{x}}_i = m_i \mathbf{v}_i + \mathbf{e}_i^u + \mathbf{e}_i^T. \quad (22)$$

Notice that if $\mathbf{e}_i^T, \mathbf{e}_i^u \rightarrow 0$, which will be later discussed, (22) becomes

$$\ddot{\mathbf{x}}_i = \mathbf{v}_i. \quad (23)$$

Then, to get good behavior, it is proposed that

$$\mathbf{v}_i = \ddot{\mathbf{x}}_{id} + \mathbf{K}_{p_i} \tilde{\mathbf{x}}_i + \mathbf{K}_{d_i} \dot{\tilde{\mathbf{x}}}_i, \quad (24)$$

where $\ddot{\mathbf{x}}_{id}$ is the desired acceleration of the trajectory, $\mathbf{K}_{p_i}, \mathbf{K}_{d_i} > 0$ are diagonal gain matrices and the position and velocity errors $\tilde{\mathbf{x}}_i$ and $\dot{\tilde{\mathbf{x}}}_i$, respectively, are defined as

$$\tilde{\mathbf{x}}_i = \mathbf{x}_{0d} - l_i \boldsymbol{\alpha}_{id} - \mathbf{x}_i, \quad (25)$$

and

$$\dot{\tilde{\mathbf{x}}}_i = \dot{\mathbf{x}}_{0d} - l_i \dot{\boldsymbol{\alpha}}_{id} - \dot{\mathbf{x}}_i. \quad (26)$$

Introducing (24) into (23), one gets

$$\ddot{\tilde{\mathbf{x}}}_i + \mathbf{K}_{d_i} \dot{\tilde{\mathbf{x}}}_i + \mathbf{K}_{p_i} \tilde{\mathbf{x}}_i = \mathbf{0}, \quad (27)$$

from which it becomes clear that the position and velocity errors converge asymptotically to zero, which indicates that the UAVs converge to their desired positions.

C. ATTITUDE CONTROLLER

To control the attitude of the aerial vehicles, it is crucial to define the desired angle values for each drone. These desired values are related to $\mathbf{u}_i$, in such a way that the z -axis of the UAV coincides with the direction of $\mathbf{u}_i$.

For the application discussed in this work, note that all movements in the drone's yaw orientation would only twist the cables; therefore, the yaw is considered to remain constant throughout the movement. Pitch and roll, on the other hand, are very closely related to the direction of each drone's thrust force.

Thus, to achieve the desired attitude, the desired pitch and roll angles of each drone, considering $\psi_i = 0$, are written as

$$\phi_{id}^0 = -\arcsin \frac{u_{iy}}{\|\mathbf{u}_i\|}, \quad (28)$$

$$\theta_{id}^0 = \left(\frac{\pi}{2} - \left| \arctan \frac{u_{iz}}{u_{ix}} \right| \right) \text{sgn}(\arctan \frac{u_{iz}}{u_{ix}}), \quad (29)$$

where sgn is the sign function, which returns 1 if its argument is positive and -1 if its argument is negative (0 otherwise) and $\|\mathbf{u}_i\|$ is the norm of the desired thrust vector. The superscript zero indicates that these values are considering the heading of the drone aligned with the inertial axis, but they can be easily modified to fit a UAV with a different yaw through the equation

$$\begin{bmatrix} \phi_{id} \\ \theta_{id} \end{bmatrix} = \begin{bmatrix} c\psi_i & s\psi_i \\ -s\psi_i & c\psi_i \end{bmatrix} \begin{bmatrix} \phi_{id}^0 \\ \theta_{id}^0 \end{bmatrix}, \quad (30)$$

where ϕ_{id} and θ_{id} are the real desired pitch and roll of each UAV, regardless of their current heading, which can be any value and will be kept constant to not twist the cables. Thus, the desired attitude vector can be written as

$$\boldsymbol{\eta}_{id} = \begin{bmatrix} \phi_{id} \\ \theta_{id} \\ \psi_{id} \end{bmatrix}. \quad (31)$$

By assuming that the drones are attached to the load via a single cable attached to their center of mass, the attitude

dynamics can be considered to be decoupled from the payload dynamics. Therefore, one can consider the control law

$$\tilde{\tau}_i = \mathbf{v}_i^\eta, \quad (32)$$

for stabilizing the decoupled attitude dynamics (6), where the desired behavior is given by

$$\mathbf{v}_i^\eta = \ddot{\eta}_{id} + \mathbf{K}_{p_i}^\eta \tilde{\eta}_i + \mathbf{K}_{d_i}^\eta \dot{\tilde{\eta}}_i, \quad (33)$$

where $\mathbf{K}_{p_i}^\eta, \mathbf{K}_{d_i}^\eta > 0$ are diagonal gain matrices and the attitude and angular speed errors $\tilde{\eta}_i$ and $\dot{\tilde{\eta}}_i$, respectively, are defined as

$$\tilde{\eta}_i = \eta_{id} - \eta_i, \quad (34)$$

and

$$\dot{\tilde{\eta}}_i = \dot{\eta}_{id} - \dot{\eta}_i. \quad (35)$$

The relationship between the controller input $\tilde{\tau}_i$ and the actual input τ_i is the inverse of (7), and is written as

$$\tau_i = \mathbf{J}\tilde{\tau}_i + \mathbf{C}(\eta, \dot{\eta})\dot{\eta}. \quad (36)$$

Therefore, it follows that

$$\ddot{\tilde{\eta}}_i + \mathbf{K}_{p_i}^\eta \tilde{\eta}_i + \mathbf{K}_{d_i}^\eta \dot{\tilde{\eta}}_i = 0. \quad (37)$$

Observe that attitude and angular velocity errors converge asymptotically to zero, indicating that the UAVs converge to their desired attitudes. In addition, since the thrust of the vehicles is an input, f_{id} can be obtained directly as long as the desired values are within the vehicle's actuator limits; thus, $\mathbf{e}_i^u \rightarrow 0$ in (22).

D. STABILITY ANALYSIS

For each vehicle, the dynamics, considering payload and cable tension disturbances, are given by (22). Here, we analyze the remaining assumption, $\mathbf{e}_i^T \rightarrow 0$ asymptotically. The closed-loop dynamics for each vehicle can be rewritten as

$$\ddot{\tilde{\mathbf{x}}}_i + \mathbf{K}_{d_i} \dot{\tilde{\mathbf{x}}}_i + \mathbf{K}_{p_i} \tilde{\mathbf{x}}_i = -\frac{1}{m_i} \mathbf{e}_i^T, \quad (38)$$

Therefore, one can propose

$$V = \frac{1}{2} \tilde{\mathbf{x}}^T \mathbf{K}_p \tilde{\mathbf{x}} + \frac{1}{2} \dot{\tilde{\mathbf{x}}}^T \dot{\tilde{\mathbf{x}}} \quad (39)$$

as the Lyapunov candidate function to analyze the stability of the i -th vehicle. Its time derivative can be written as

$$\dot{V} = \tilde{\mathbf{x}}^T \mathbf{K}_p \dot{\tilde{\mathbf{x}}} + \dot{\tilde{\mathbf{x}}}^T (-\mathbf{K}_d \dot{\tilde{\mathbf{x}}} - \mathbf{K}_p \tilde{\mathbf{x}} - \frac{1}{m} \mathbf{e}^T) \quad (40)$$

which, with some manipulation, can be simplified to

$$\begin{aligned} \dot{V} &= -\dot{\tilde{\mathbf{x}}}^T \mathbf{K}_d \dot{\tilde{\mathbf{x}}} - \frac{1}{m} \dot{\tilde{\mathbf{x}}}^T \mathbf{e}^T \\ &\leq -\lambda_{\min}(\mathbf{K}_d) \|\dot{\tilde{\mathbf{x}}}\|^2 + \frac{1}{m} \|\dot{\tilde{\mathbf{x}}}\| \|\mathbf{e}^T\| \end{aligned}$$

from where a sufficient condition to guarantee that $\dot{V}$ is negative semi-definite is $\|\dot{\tilde{\mathbf{x}}}\| \geq \frac{\delta/m}{\lambda_{\min}(\mathbf{K}_d)} \in \Omega$.

Since $\dot{V}$ is negative semi-definite in Ω , V is bounded, thus, trajectories starting in $\|\dot{\tilde{\mathbf{x}}}\| < \frac{\delta/m}{\lambda_{\min}(\mathbf{K}_d)}$, i.e., in the complement of Ω , are also bounded. In addition, since the system is autonomous and the inequalities mentioned above hold for $t \geq t_0$, Ω is an invariant set, and, according to LaSalle Invariance Principle, every trajectory starting in Ω approaches the set $M = \{(\tilde{\mathbf{x}}, \dot{\tilde{\mathbf{x}}}) : \dot{V} = 0\} \subseteq \Omega$ as $t \rightarrow \infty$.

Although $\dot{V}$ is equal to zero in $\dot{\tilde{\mathbf{x}}} = \mathbf{0}$ and $\dot{\tilde{\mathbf{x}}} = -\frac{1}{m} \mathbf{K}_d^{-1} \mathbf{e}^T$, one can conclude that the later solution does not stay in M . Therefore, considering the solution $\dot{\tilde{\mathbf{x}}} = \mathbf{0}$ in the system dynamics of (38), we conclude that the largest invariant set is given by $M = \left[\begin{smallmatrix} \tilde{\mathbf{x}} \\ \dot{\tilde{\mathbf{x}}} \end{smallmatrix} \right] = \left[\begin{smallmatrix} -\frac{1}{m} \mathbf{K}_p^{-1} \mathbf{e}^T \\ \mathbf{0} \end{smallmatrix} \right]$. Finally, one can conclude that since $\|\mathbf{e}^T\|$ is bounded by δ , the error dynamics for each quadrotor is *ultimately bounded*. Moreover, since $\mathbf{K}_p$ and $\mathbf{K}_d$ can be arbitrarily chosen, the convergence bound can be chosen to be conveniently small. Therefore, the assumption of asymptotic convergence of $\mathbf{e}_i^T \rightarrow 0$ holds, in this case, configuring what is called practical stability in the literature [32].

IV. MODIFICATION OF THE NUMBER OF AGENTS

The load controller shown in the last section was developed to work for n agents. This fact provides the flexibility to form a team with any number of robots as the task requires. The discussion assumed that, once the number of agents was decided, it would remain static throughout the task. However, some tasks, such as navigating narrow corridors, obstacle avoidance, or the failure of one of the agents, may require or benefit from dynamically adjusting the number of agents while the task is being carried out.

Therefore, in our analysis, we consider modifying the number of UAVs that carry the load. To accomplish this, the premise that the cables are rigid is relaxed. This makes practical sense, as the cables used are usually wire-like materials that are pretty close to rigid when under tension from the load, but can bend freely when this tension is removed. Therefore, the method by which this work alters the number of agents in the formation is simply by causing the desired removed vehicles to move closer to the load. This eliminates the tension force on the respective cable, thus eliminating its contribution to the virtual controller moving the load and redistributing the weight to the other agents.

This, of course, if done carelessly, may introduce oscillation in the system. That is why, if possible, a reconfiguration of the current geometric configuration of the formation is carried out to try to eliminate most of the agent-to-be-removed influence on the load prior to the approach maneuver. For example, consider the scenario illustrated in Fig. 1, where three UAVs are transporting a load, and UAV 3 is to be removed. The algorithm first adjusts the positions of the other UAVs—in this case, moving UAV 2 so that it becomes aligned with UAV 1 in a plane defined by $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2$. Simultaneously, UAV 3 moves to a position perpendicular to this plane. This maneuver reduces UAV 3's contribution to

the net force on the load, allowing for a smoother detachment. The resulting configuration is depicted in Fig. 3.

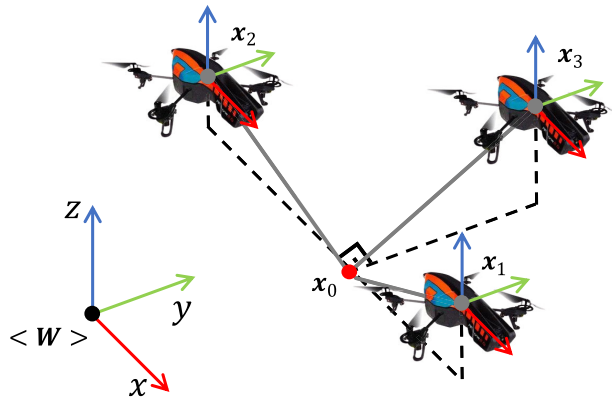


FIGURE 3. Graphical representation of the system agents and the configuration variables for $n = 3$ preparing to change to $n = 2$.

This change virtually eliminates the influence of the UAV 3 on the load, redistributing its weight to the cables of the other two agents. Then, the third agent can move closer to the load with minimal impact on the system.

A question remains, however: How should an agent that left its position in the formation position itself? It must respect the system's physical restrictions, of course, such as maintaining a safe distance from the load within the length of its cable and avoiding interaction with other cables. Other than that, however, these agents are free to position themselves in various ways. To ensure that, this work proposes a leader-follower structure between each free agent and the load.

A. CONTROLLER OF THE FREE AGENTS

Some ideas from [33] are taken for the leader-follower structure. It is used here to ensure that each free agent can reach and maintain a relative position to the load. As each free agent is a separate follower, from now on this work will deal with a single free agent, which has position $\mathbf{x}_f$ and orientation η_f , where the subscript f refers to its role as a follower.

The desired position for the free agent is defined as

$$\mathbf{x}_{fd} = \mathbf{x}_0 + \mathbf{R}_l \mathbf{x}_{off}, \quad (41)$$

where $\mathbf{x}_0$ is the current position of the load, acting as the leader, $\mathbf{x}_{off}$ is a vector with the Cartesian desired offset between the leader and follower and $\mathbf{R}_l$ is a rotation matrix defined as

$$\mathbf{R}_l = \begin{bmatrix} c_{\psi_l} & -s_{\psi_l} & 0 \\ s_{\psi_l} & c_{\psi_l} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (42)$$

where ψ_l is a pseudo orientation for the load and the subscript l refers to its role as related to the leader. The reason for this pseudo-orientation is that it may be easier to position the free agent relative to the load, regardless of the positions of the rest

of the formation, and it is chosen depending on the geometric configuration of the system.

As an example, consider the situation depicted in Fig. 3, where ψ_l is the angle of the plane passing through $\mathbf{x}_0$, $\mathbf{x}_1$ and $\mathbf{x}_2$ in the inertial z -axis. If one were to place the free agent a fixed distance to the left of the load, where it is free of any other cables, and then rotate the whole system in a way that the load stays still, but the UAVs 1 and 2 describe a circle around the z -axis, the controller with the pseudo-orientation would make sure that the free agent would navigate to the relative position to the left of the load, describing a smaller circle around the load, which will be free of the cables of the other agents.

Now, in order to find the desired velocity, (41) is differentiated, yielding to

$$\dot{\mathbf{x}}_{fd} = \dot{\mathbf{x}}_0 + \mathbf{R}_l \dot{\mathbf{x}}_{off} + \dot{\mathbf{R}}_l \mathbf{x}_{off}. \quad (43)$$

Then, consider the model of the follower shown in (3) without the influence of the tension force, or

$$m_f \ddot{\mathbf{x}}_f = f_f \mathbf{R}(\eta_f) \mathbf{e}_3 - m_f g \mathbf{e}_3, \quad (44)$$

and, by expanding it, it is possible to write

$$m_f \ddot{x}_f = f_f (\sin \phi_f \sin \psi_f + \cos \phi_f \sin \theta_f \cos \psi_f) \quad (45)$$

$$m_f \ddot{y}_f = f_f (\cos \phi_f \sin \theta_f \sin \psi_f - \sin \phi_f \cos \psi_f) \quad (46)$$

$$m_f \ddot{z}_f = f_f \cos \phi_f \cos \theta_f - m_f g, \quad (47)$$

which is the expanded translational model of the follower robot.

Now, consider the control law

$$f_f = \frac{m_f (v_z + g)}{\cos \phi_f \cos \theta_f}, \quad (48)$$

where

$$v_z = \ddot{z}_{fd} + k_{d_z}^f \dot{\tilde{z}}_f + k_{p_z}^f \tilde{z}_f, \quad (49)$$

for the thrust, where $k_{d_z}^f$ and $k_{p_z}^f$ are positive constants, and $\tilde{z}_f$, $\dot{\tilde{z}}_f$ are the position and velocity errors, respectively, defined as $\tilde{z}_f = z_{fd} - z_f$ and $\dot{\tilde{z}}_f = \dot{z}_{fd} - \dot{z}_f$.

By introducing (48)-(49), one gets

$$\ddot{x}_f = (v_z + g) \left(\frac{\tan \phi_f}{\cos \theta_f} \sin \psi_f + \tan \theta_f \cos \psi_f \right), \quad (50)$$

$$\ddot{y}_f = (v_z + g) \left(\tan \theta_f \sin \psi_f - \frac{\tan \phi_f}{\cos \theta_f} \cos \psi_f \right), \quad (51)$$

$$\ddot{z} = \ddot{z}_{fd} + k_{d_z}^f \dot{\tilde{z}}_f + k_{p_z}^f \tilde{z}_f, \quad (52)$$

and by choosing the desired roll and pitch angles as

$$\phi_d = \arctan \frac{(v_x \sin \psi_f - v_y \cos \psi_f) \cos \theta_f}{v_z + g}, \quad (53)$$

$$\theta_d = \arctan \frac{(v_x \cos \psi_f + v_y \sin \psi_f)}{v_z + g}, \quad (54)$$

where

$$v_x = \ddot{x}_{fd} + k_{d_x}^f \dot{\tilde{x}}_f + k_{p_x}^f \tilde{x}_f \quad (55)$$

and

$$v_y = \ddot{y}_{fd} + k_{d_y}^f \dot{\tilde{y}}_f + k_{p_y}^f \tilde{y}_f, \quad (56)$$

equations (50)-(52) become

$$\ddot{\tilde{x}}_f + k_{d_x} \dot{\tilde{x}}_f + k_{p_x} \tilde{x}_f = 0, \quad (57)$$

$$\ddot{\tilde{y}}_f + k_{d_y} \dot{\tilde{y}}_f + k_{p_y} \tilde{y}_f = 0, \quad (58)$$

$$\ddot{\tilde{z}}_f + k_{d_z} \dot{\tilde{z}}_f + k_{p_z} \tilde{z}_f = 0, \quad (59)$$

which shows that the position and velocity errors converge asymptotically to zero when time converges to infinity. Ultimately, this means that the follower will reach its desired relative position to the leader, i.e., the free agent will position itself in a desired position in space, relative to the load.

All that remains is to ensure that the follower free-agent reaches the desired angle values. This is accomplished through the same control law as in section III-C. The difference is that the desired values of roll and pitch angles are now given by (53) and (54), while the desired yaw value is free.

V. SIMULATIONS

This section describes the simulations conducted to validate the approach presented in the previous sections. To do so, three situations were devised, the first one showing the load controller alone, following a circular trajectory, while the second one follows a line trajectory with the modification of the number of agents, and the last one is again of a circular motion, but this time with 5 UAVs executing the tasks, showing the scalability of the proposed controller. These simulations were performed using the software *Matlab* and their results are shown in the following subsections.

A. SIMULATION #1: CIRCULAR TRAJECTORY

As previously mentioned, this simulation shows the system following a circular trajectory. The chosen trajectory is described by

$$\mathbf{x}_{0d} = \begin{bmatrix} 1 - \cos \omega t \\ \sin \omega t \\ \frac{1}{2} - \frac{1}{2} \cos \omega t \end{bmatrix}, \quad (60)$$

where t is the time elapsed since the beginning of the simulation, and $\omega = \frac{2\pi}{T}$, where $T = 20s$ is the period of the trajectory.

The desired tensions can be generated by (10) and (14). However, this would result in a system with an infinite number of solutions; therefore, additional constraints are necessary to select one solution. Despite several other possibilities, our choice was to consider the desired tensions for the second and third robots as constants. This results in a system where the first robot acts as the agent of movement while the other agents act as more stabilizing forces. With that

in mind, the desired tensions are defined as

$$T_{1d} \alpha_{1d} = -\mathbf{u}_0 - T_{2d} \alpha_{2d} - T_{3d} \alpha_{3d}, \quad (61)$$

$$T_{2d} \alpha_{2d} = -\frac{m_0}{3} g \mathbf{R}(0, -\pi/6, -\pi/4) \mathbf{e}_3, \quad (62)$$

$$T_{3d} \alpha_{3d} = -\frac{m_0}{3} g \mathbf{R}(0, -\pi/6, \pi/4) \mathbf{e}_3. \quad (63)$$

As for the initial positions, they are defined as $\mathbf{x}_0 = [0 \ 0 \ 0]^T$ for the load and

$$\mathbf{x}_1 = l_1 \mathbf{R}(0, \pi/6, 0) [0 \ 0 \ 1]^T, \quad (64)$$

$$\mathbf{x}_2 = l_2 \mathbf{R}(0, -\pi/6, \pi/4) [0 \ 0 \ 1]^T, \quad (65)$$

$$\mathbf{x}_3 = l_3 \mathbf{R}(0, -\pi/6, -\pi/4) [0 \ 0 \ 1]^T, \quad (66)$$

for the drones. In section III-A, it was said that the initial conditions must be well posed, and indeed, by comparing (62)-(63) to (65)-(66) it is possible to see that the second and third agents are already at the desired tension angle. Furthermore, it is possible to observe that when $t = 0$, Eq. (72) is equal to $\mathbf{x}_0$, meaning that the load is already positioned at the beginning of the trajectory. Moreover, (64)-(66) follow the template of (2), meaning that all drones are already correctly positioned around the load, with the cables taut.

Finally, the remaining initial conditions are

$$m_0 = 0.139 \text{ Kg},$$

$$m_i = 0.5 \text{ Kg},$$

$$l_i = 1 \text{ m},$$

$$\mathbf{K}_{p0} = \text{diag} [0.075 \ 0.075 \ 0.075],$$

$$\mathbf{K}_{d0} = \text{diag} [0.03 \ 0.03 \ 0.03],$$

$$\mathbf{K}_{p_i} = \text{diag} [2 \ 2 \ 2],$$

$$\mathbf{K}_{d_i} = \text{diag} [1 \ 1 \ 1],$$

$$\mathbf{K}_{p_i}^\eta = \text{diag} [5 \ 5 \ 5],$$

$$\mathbf{K}_{d_i}^\eta = \text{diag} [3 \ 3 \ 3].$$

The complete navigation of the system can be watched at <https://youtu.be/BKxDFHOPgAI>, while a snapshot of the navigation can be seen in Fig. 4. In this figure, it is possible to observe that the load, represented in solid black, converges to its desired position, depicted in dashed black. It is also possible to observe the transient at the beginning of the movement, where the load had to accelerate to align with the desired trajectory.

The time evolution of the load errors is represented in Fig. 5. Regarding the x - and z -axes, we have maximum errors in the order of two to three centimeters, whereas the maximum error in the y -axis is in the order of twenty centimeters. This occurs due to the intrinsic characteristics of the trajectory, as it requires maximum velocity in y at $t = 0$ and zero velocity for x and z . Nevertheless, all errors converge to values close to zero, as shown in the smaller rectangles, which depict the zoomed-in errors after the system reaches the trajectory and the transient is over.

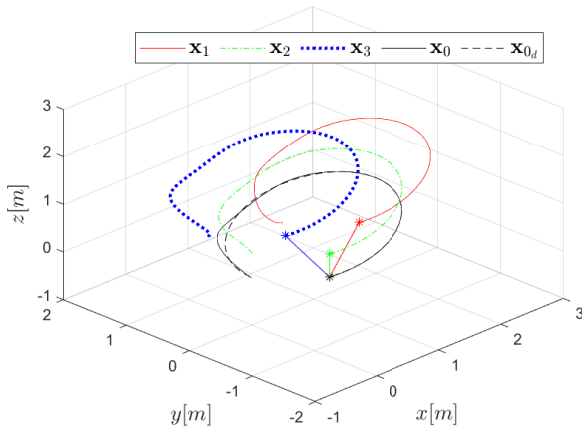


FIGURE 4. Depiction of the navigation in Simulation #1.

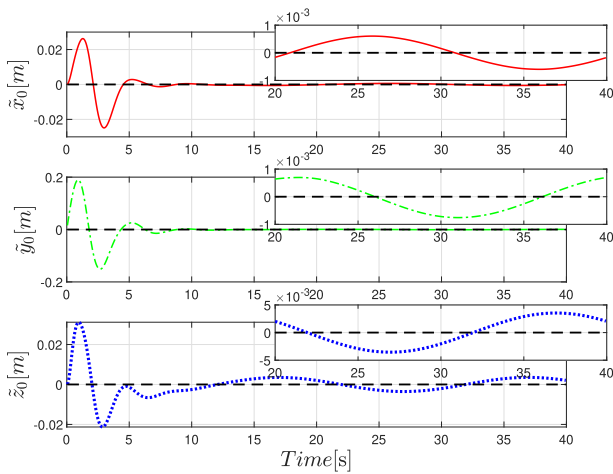


FIGURE 5. Performance of the load errors in Simulation #1.

Continuing with the controllers, Fig. 6 illustrates the tension forces applied to each agent. Here, the actual and desired tension forces are shown, rather than their difference, which is the error. This is done purposely to illustrate that the desired tension forces in the agents 2 and 3 remain constant throughout the movement, while the desired tension forces in the first agent change to create the desired movement. This is consistent with equations (61)–(63). Furthermore, it is possible to observe that all the tension values converge to their desired values.

Finally, Fig. 7 shows the roll and pitch errors. The yaw error was omitted to avoid cluttering since the heading of the drones has little impact on the movements being observed. It is possible to see that, much like the load errors, the errors quickly converge to values near zero after the first few seconds. Again, the smaller boxes show the magnitude of the errors after the transient is over, around 0.01 degrees.

Thus, the video and the graphics shown allow the conclusion that the load effectively tracks the desired trajectory, in accordance with the theory presented so far in this work.

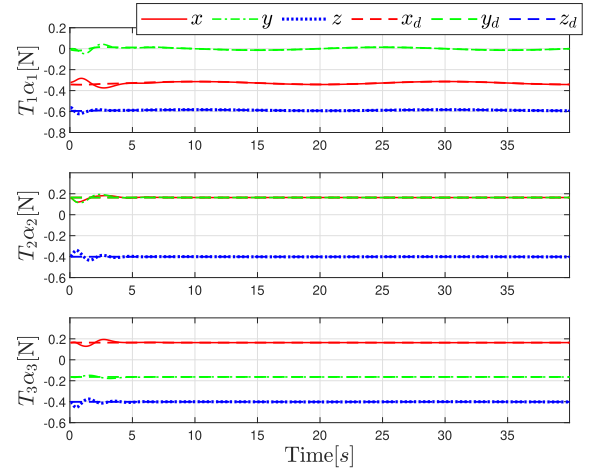


FIGURE 6. Performance of the actual and desired tension forces in Simulation #1.

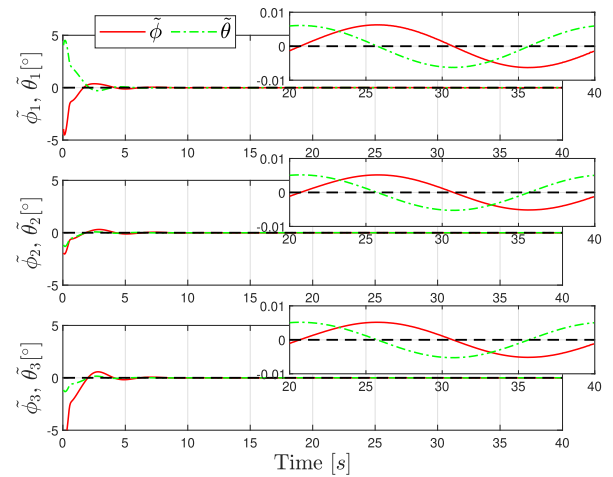


FIGURE 7. Behavior of the attitude errors in Simulation #1.

B. SIMULATION #2: LINE TRAJECTORY CHANGING THE NUMBER OF AGENTS

This simulation was made to show the modification of the number of agents. To showcase its capabilities, a line trajectory was chosen to emulate a task that would require the formation to navigate through a narrow corridor. The task is divided into three phases: in the first phase, the formation will initiate movement in the same configuration as in the previous simulation. The second phase is the transition phase, where the second and third agents reposition themselves, as described in Section IV. In the final phase, since its influence has been removed, the third agent will move closer to the load, resulting in a much smaller formation in the y direction, which enables its navigation through narrow corridors.

The chosen trajectory is described by

$$\mathbf{x}_{0d} = \begin{bmatrix} t \\ 10 \\ 0 \\ 0 \end{bmatrix}, \quad (67)$$

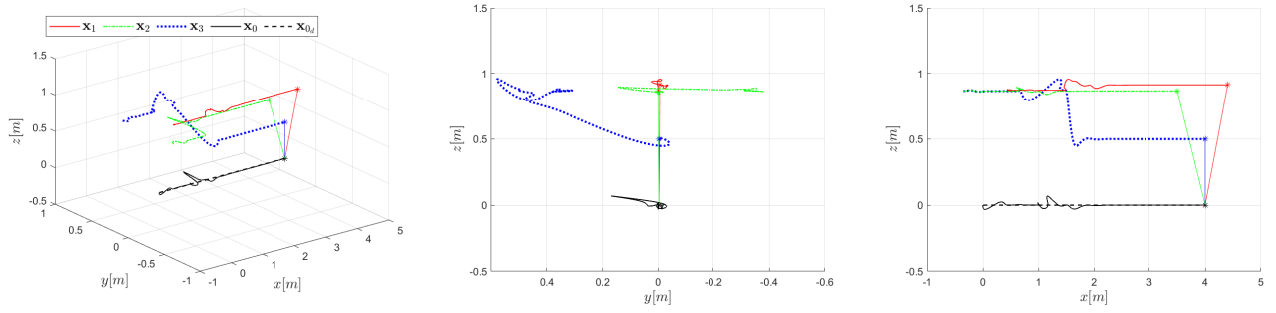


FIGURE 8. Navigation of the system in Simulation #2, shown in three different views: aerial, rear, and lateral.

and the phase transitions happen when $t = 10s$ and $t = 15s$. The desired tensions for the first phase are the same as described in (61)-(63), and for the second phase, they are defined as

$$T_{1d}\alpha_{1d} = -\mathbf{u}_0 - T_{2d}\alpha_{2d}, \quad (68)$$

$$T_{2d}\alpha_{2d} = -\frac{m_0}{2}g\mathbf{R}(0, -\pi/6, 0)\mathbf{e}_3, \quad (69)$$

$$T_{3d}\alpha_{3d} = -\frac{m_0}{10}g\mathbf{R}(0, -\pi/6, \pi/2)\mathbf{e}_3. \quad (70)$$

Although similar, there are a few notable differences in these new values: the first one in (68) is that the influence of the third agent has been removed from the first agent, i.e., the one who actively creates the movement. The second difference is in (69), where before it had a 45° angle in the z -axis, now it has a 0° angle, meaning it will position itself directly behind the first agent. Another difference related to the last one is that the desired angle in the z -axis of the third agent is now 90° , meaning that it will position itself at the side of the formation. With these changes, the formation will change from the one depicted in Fig. 1 to the one shown in Fig. 3. Lastly, the magnitude of the desired tension has been changed from one-third to each agent, to one-half for agents 1 and 2, and to one-tenth for agent 3 (the desired tension should be zero in the last agent, but to transition it using the same controller it has been found that it is necessary to have a small contribution in the desired tension).

For the last phase, the desired tensions described in (68)-(69) remain, and as the last agent uses a different controller, it does not have a desired or actual tension value anymore. Instead, it seeks to keep an offset value with respect to the load, defined as

$$\mathbf{x}_{off} = \begin{bmatrix} 0 \\ 0 \\ 0.5 \end{bmatrix}. \quad (71)$$

This means that the desired position of the third agent is directly on top of the load, at the height of 50 centimeters above it. The initial conditions are the same as those in the last simulation, except for the gains of the free agent controller, which were not present in the previous example. These gains

are defined as

$$\mathbf{K}_{p_z}^f = \text{diag}[5 \quad 5 \quad 5],$$

$$\mathbf{K}_{d_z}^f = \text{diag}[3 \quad 3 \quad 3].$$

The simulation can be watched at https://youtu.be/_3gZoJvbnBI, while three views of the movement can be seen in Fig. 8. These different perspectives of the simulation show that the free agent could position itself in the middle of the other agents, reducing the system's width. Indeed, upon examining the view from the back, one can observe that by the end of the movement, the formation profile has been reduced to positions around $y = 0$. Furthermore, looking at the lateral view, it is shown that the free agent has positioned itself correctly between the others.

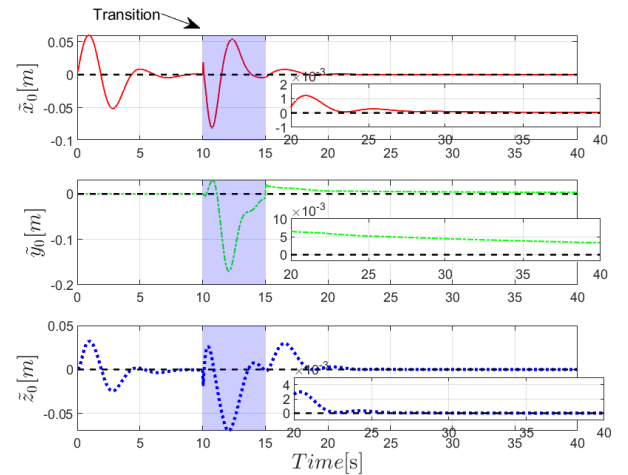


FIGURE 9. Performance of the load errors in Simulation #2.

The evolution in time of the load errors is presented in Fig. 9. A solid rectangle marks the transitioning period, while rectangular boxes show a zoomed-in view of the errors after the system has reached a steady state. Due to the nature of the trajectory, which has velocity pointing only in the x -axis direction, we observe a maximum initial error of around 5 centimeters, which is quickly reduced to values near zero. Meanwhile, on the y -axis, the initial error is quite small. Interestingly, even though there was no change in the desired

height, the initial acceleration caused an error in z , but that was also quickly reduced.

Once the transition phase begins, the errors increase, but they are quickly controlled and decrease to zero after the transition period finishes. The error in y is around 20 centimeters due to the oscillation of the load. Finally, all errors are quickly reduced to really small values after the transition period, as the zoomed-in boxes show. Interestingly, the graph in y shows a steady error. This is probably caused by the fact that some DOFs were lost when changing the formation to two agents, and now the cable's tension forces cannot freely control the y -axis direction perpendicular to their plane.

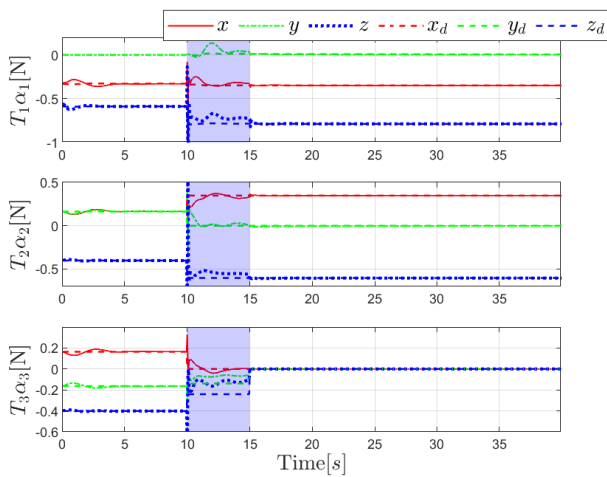


FIGURE 10. Performance of the actual and desired tension forces in Simulation #2.

Figure 10 shows the tension values in each agent, and again, the blue rectangle marks the transitioning period. It is perceivable that after the transition begins, the desired tensions for the agents 1 and 2 increase accordingly, and the actual tension values follow them, enabling the load to continue following the trajectory. Conversely, the desired tension for the third agent decreases and ultimately approaches zero, as it no longer affects its movement or formation.

Finally, Fig. 11 shows the attitude errors in each agent. This graph illustrates the distinct roles of each agent across the different phases. For the first agent, its role in the entire movement remains unchanged, as reflected in the small errors shown. However, the second and third agents move their relative positions to the load once the transition period begins, reflected in an increase in errors around the 10-second mark. Finally, after the transition period ends, the third agent moves to the middle, which changes its desired position suddenly, causing the errors to spike around the 15-second mark. Nonetheless, throughout all the movements, the attitude errors are always quickly reduced to values near zero and remain there, as the zoomed-in boxes show.

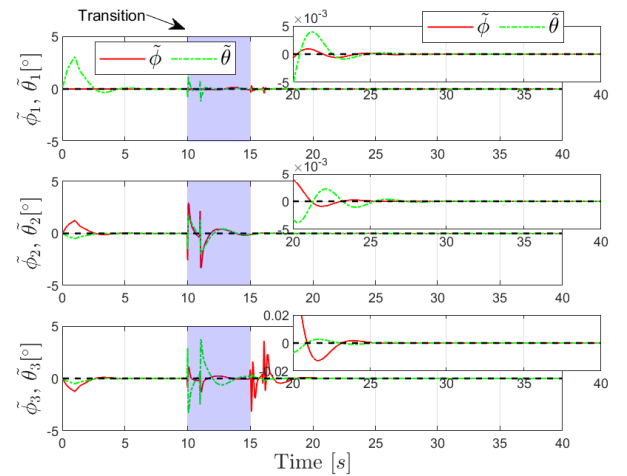


FIGURE 11. Behavior of the attitude errors in Simulation #2.

This simulation has demonstrated the system's capability to make the load follow a trajectory, even when the number of agents responsible for its movement changes.

C. SIMULATION #3: CIRCULAR TRAJECTORY WITH 5 UAVS

This last simulation aims to showcase the scalability of the proposed method. In order to do so, the desired trajectory is described by

$$\mathbf{x}_{0d} = \begin{bmatrix} 1 - \cos \omega t \\ \sin \omega t \\ \frac{1}{2} - \frac{1}{2} \cos \omega t \end{bmatrix}, \quad (72)$$

which is the same as that adopted in the first simulation, including the same period of 20 seconds.

The desired tensions, once again, need more care to be chosen. There are infinite solutions, and some constraints must be put in place to choose one of them. For this simulation, the chosen constraints aimed to maintain constant desired tensions for the second, third, fourth, and fifth UAVs. Meanwhile, the first robot would move to create the desired force on the load, acting once again as the agent of movement. Thus, the desired tensions assume the form of

$$T_{1d}\alpha_{1d} = -\mathbf{u}_0 - \sum_{i=2}^5 T_{id}\alpha_{id}, \quad (73)$$

$$T_{2d}\alpha_{2d} = -\frac{m_0}{5}g\mathbf{R}(0, -\pi/6, \pi/2)\mathbf{e}_3, \quad (74)$$

$$T_{3d}\alpha_{3d} = -\frac{m_0}{5}g\mathbf{R}(0, -\pi/6, -\pi/6)\mathbf{e}_3, \quad (75)$$

$$T_{4d}\alpha_{4d} = -\frac{m_0}{5}g\mathbf{R}(0, \pi/6, \pi/2)\mathbf{e}_3, \quad (76)$$

$$T_{5d}\alpha_{5d} = -\frac{m_0}{5}g\mathbf{R}(0, -\pi/6, \pi/6)\mathbf{e}_3. \quad (77)$$

Meanwhile, the initial positions are defined as $\mathbf{x}_0 = [0 \ 0 \ 0]^T$ for the load and

$$\mathbf{x}_1 = l_1\mathbf{R}(0, \pi/6, 0)[0 \ 0 \ 1]^T, \quad (78)$$

$$\mathbf{x}_2 = l_2 \mathbf{R}(0, -\pi/6, \pi/2) [0 \ 0 \ 1]^T, \quad (79)$$

$$\mathbf{x}_3 = l_3 \mathbf{R}(0, -\pi/6, -\pi/6) [0 \ 0 \ 1]^T, \quad (80)$$

$$\mathbf{x}_4 = l_4 \mathbf{R}(0, \pi/6, \pi/2) [0 \ 0 \ 1]^T, \quad (81)$$

$$\mathbf{x}_5 = l_5 \mathbf{R}(0, -\pi/6, \pi/6) [0 \ 0 \ 1]^T, \quad (82)$$

for the drones. Finally, the remaining initial conditions are the same as those of the last simulations, with the exception of the cable length, which is defined as $l_i = 1.5m$. This was changed to accommodate the increased number of robots, and to allow better visualization.

The complete simulation can be watched at https://youtu.be/XK9Bqhrn_6A, while a snapshot of the navigation is shown in Fig. 12. The Figure shows that, similar to the simulation with only three UAVs, the load converges to its desired position after an initial transient, during which it accelerates to align with its desired trajectory.

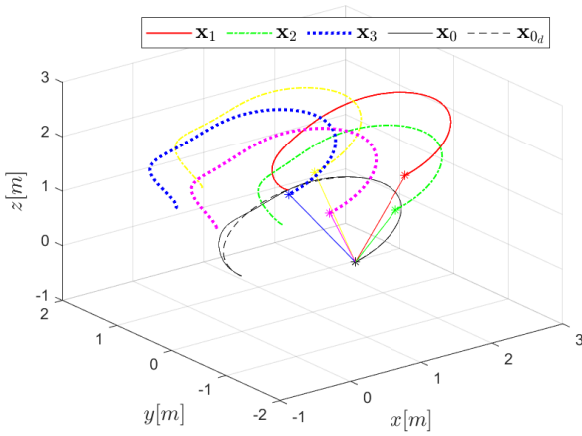


FIGURE 12. Depiction of the navigation in Simulation #3.

The time evolution of the load errors is depicted in Fig. 13. The errors on the x - and z -axes, in the order of 5 centimeters, are once again smaller than the errors on the y -axis, which are on the order of 20 centimeters. As previously explained, this happens because the selected desired trajectory demands maximum acceleration on the y -axis. Nonetheless, all errors are soon controlled and converge to values near zero, as the smaller rectangles depicting the zoomed-in errors after the transient show. Around $t = 35$ seconds, a small error spike occurs, which is soon controlled, likely due to a stutter in the simulation.

As for the tension forces in each agent, they are shown in Figure 14 and 15. As expected, and consistent with equations (73)–(75), the desired tensions in agents 2 through 5 remain constant, while the desired tensions in agent 1 create the desired movement of the load.

The roll and pitch errors are presented in Fig. 16 and 17, where it can be seen that they start with large values, due to the initial acceleration, but quickly converge to values near zero, as seen in the smaller rectangular boxes. Moreover, the spike that can be seen in Fig. 13, depicting the load errors,

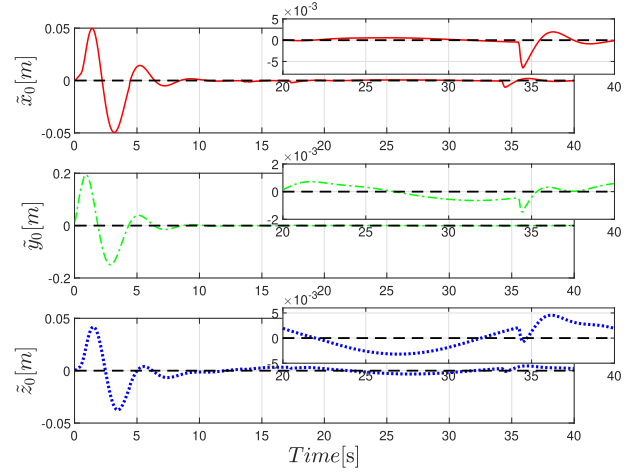


FIGURE 13. Performance of the load errors in Simulation #3.

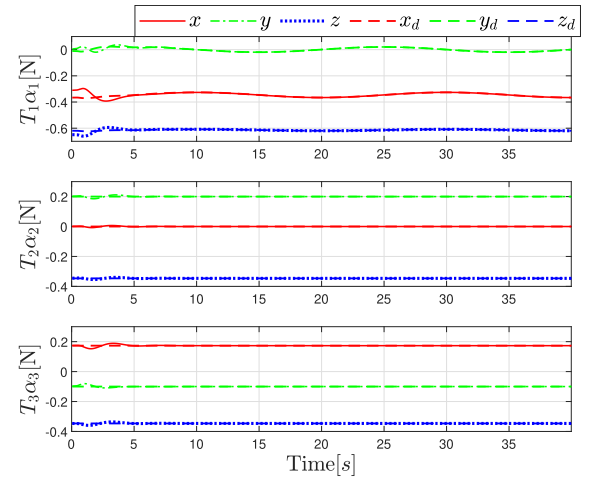


FIGURE 14. Performance of the actual and desired tension forces of agents 1, 2, and 3 in Simulation #3.

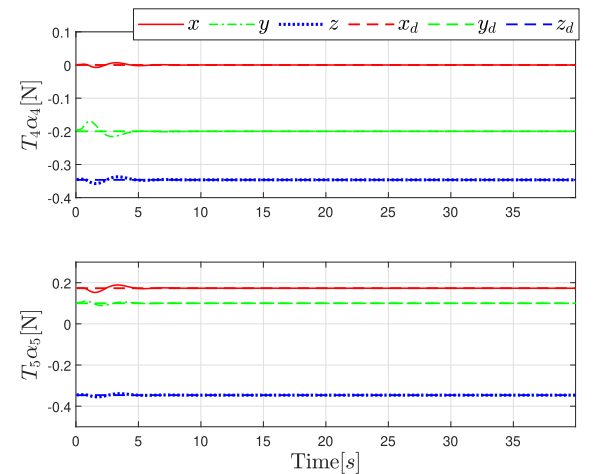


FIGURE 15. Performance of the actual and desired tension forces of agents 4 and 5 in Simulation #3.

is also evident here, as the robots must momentarily increase their accelerations to compensate for the disturbance.

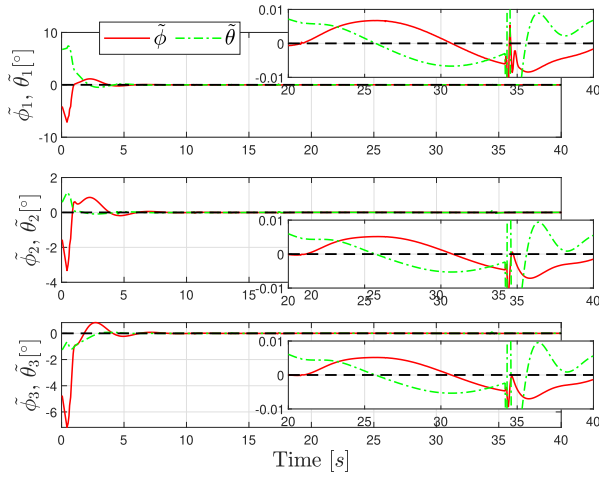


FIGURE 16. Behavior of the attitude errors of agents 1, 2, and 3 in Simulation #3.

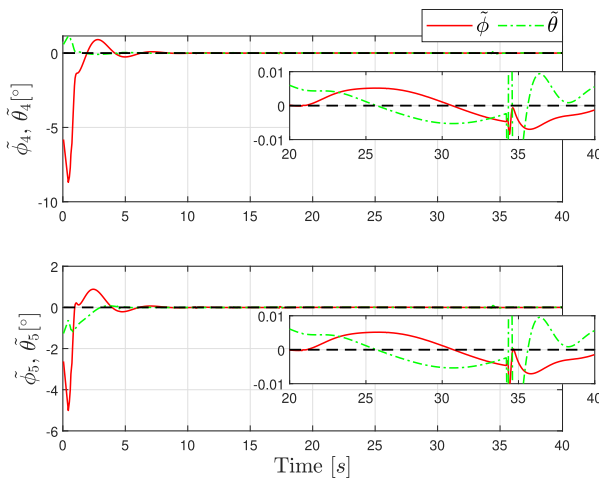


FIGURE 17. Behavior of the attitude errors of agents 4 and 5 in Simulation #3.

Finally, the video and analysis of the figures show that the trajectory tracking goals were achieved, even with an increase in the number of agents. Thus, this simulation demonstrates the scalability of the proposed solution, provided suitable initial conditions and tension constraints are chosen.

VI. EXPERIMENTS

This section describes the real-world experiments conducted to validate the work and the simulations presented so far. To achieve that, two experiments similar to the simulations shown were done, and their results are presented in this section. Furthermore, a last experiment is presented, where heavy external disturbance is introduced into the system by a pole to test the system's stability.

These experiments were conducted at the *Heudiasyc* laboratory in Compiègne, France. The drones used were *AR Drones* from *Parrot*. The position feedback of the objects was obtained via the motion capture system *OptiTrack*, and the mass of the load and the agents was determined through

the use of a scale. FL-Air (Framework libre AIR) [34] was used to control the whole system, which is responsible for managing each robot and the information from the motion capture system.

In these experiments, a modification was necessary to accommodate the system's requirements. In simulation, the drones can start already in the air with the cable taut, but this is impossible in an experiment. Because of this, a take-off in a safe mode was used. This means the drones will take off using their embedded controller, ignoring the load, and position themselves directly on top of their takeoff positions at a pre-configured height. Only after the drones reach their desired height and are stable with the load already in the air and the cable taut does the load controller take charge, and the experiment begins. For this to work, the drones are positioned so that the cables are stretched, and their positions are close to the initial positions required by the experiment, thereby maintaining the validity of the well-posed initial conditions. The videos presented in each section effectively illustrate this situation.

A. EXPERIMENT #1: CIRCULAR TRAJECTORY

This first experiment closely follows the simulation of tracking a circular trajectory of the previous section. However, since the safe takeoff will leave the load in an uncertain position, the desired trajectory was changed to reflect this, now being described by

$$\mathbf{x}_{0d} = \begin{bmatrix} 1 - \cos \omega t \\ \sin \omega t \\ \frac{1}{2}(1 - \cos \omega t) \end{bmatrix} + \mathbf{x}_0^{st}, \quad (83)$$

where $\mathbf{x}_0^{st}$ is the current position of the load once the safe take-off finishes and the load controller takes over. This means that for $t = 0$, $\mathbf{x}_{0d} = \mathbf{x}_0^{st}$, which means that the trajectory starts at the current position of the load, avoiding large initial accelerations to be required by the trajectory. The trajectory period is 20 seconds, as in the simulation.

As for the desired tensions, they are equal to the ones in (61)-(63), and the remaining initial conditions are

$$m_0 = 0.139 \text{ Kg},$$

$$m_i = 0.5 \text{ Kg},$$

$$l_i = 1.5 \text{ m},$$

$$\mathbf{K}_{p0} = \text{diag} [1 \quad 1 \quad 0.5],$$

$$\mathbf{K}_{d0} = \text{diag} [0.2 \quad 0.2 \quad 0.2],$$

$$\mathbf{K}_{pi} = \text{diag} [0.5 \quad 0.5 \quad 0.5],$$

$$\mathbf{K}_{di} = \text{diag} [0.2 \quad 0.2 \quad 0.2].$$

Notice that the gains for the attitude controller were not shown. This is because the FL-Air platform already has a well-tuned low-level controller that transforms desired attitudes into torques.

The experiment can be viewed at <https://youtu.be/TtLddVnEIOE>, where smooth navigation is demonstrated through relatively fast movement. Figure 18 shows the

complete navigation of the experiment, where it can be seen that the load successfully followed a circular trajectory, even though it shows some distance to the desired trajectory.

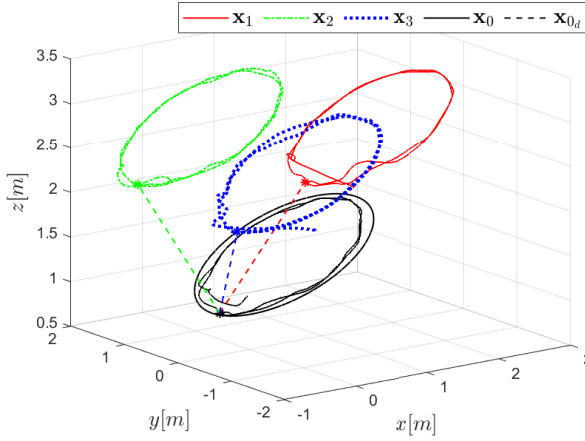


FIGURE 18. Navigation performance of Experiment #1.

The evolution of errors over time can be seen in Figure 19. As the image shows, a larger periodic error is present in the x - and y -directions, of the order of 20 centimeters. However, smaller errors in the z -direction are presented, usually less than 5 centimeters.

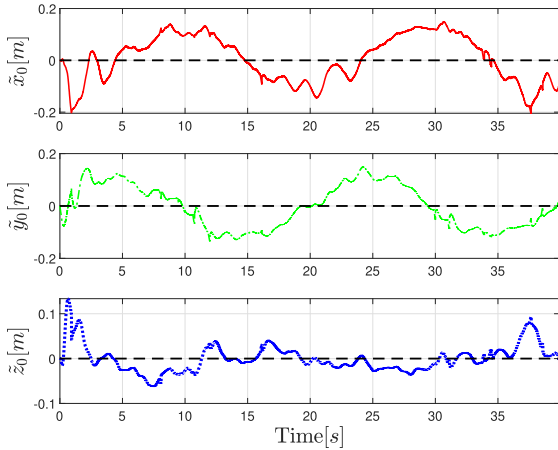


FIGURE 19. Performance of the load errors in Experiment #1.

Finally, the desired and actual tensions are shown in Fig. 20. Unlike the simulation, however, where the actual tension was a deterministic value necessary to calculate the future iterations of the simulation, in the experiments, the actual tension is never used in the controller. Due to this, the actual tension values shown in Figure 20 are estimated using the acceleration of the load, its mass, gravity, and the relative position of the drones. This, however, especially the acceleration of the load, which is not directly measured but obtained through the double differentiation of the load position provided by the motion capture system, introduces some noise in the results, making it difficult to quantify accurately.

Nevertheless, it is possible to see that the behavior presented in Figure 6 is broadly followed in Figure 20. Furthermore, the tension values exhibit minor errors compared to the desired values. This indicates that the controller achieved the desired tension values in the cables. Additionally, it can be seen that the tensions for the first agent present more significant oscillations. This is because, unlike the other agents, where the desired tensions are constant, the desired tensions for the first agent vary slightly over time. It is also the agent responsible for responding to the changes in acceleration along the trajectory.

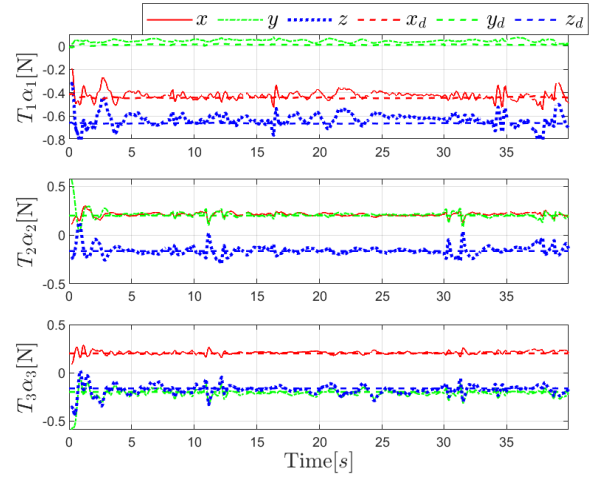


FIGURE 20. Performance of the actual and desired tension forces in Experiment #1.

Ultimately, it can be said that although errors of around 20 centimeters appear in navigation, the load tracked the general behavior expected of it. If a slower trajectory had been selected, these errors would be much smaller. However, this work aimed to demonstrate that, despite the approach requiring low accelerations, it can be applied in applications with significant travel speeds. Some future work to mitigate experimental errors and more robust options to deal with uncertainties are discussed in the conclusion of this work.

B. EXPERIMENT #2: LINE TRAJECTORY CHANGING THE NUMBER OF AGENTS

Like the first experiment in the previous section, this one also tries to closely follow the simulation to present a better contrast. However, unlike the simulation, where there is unlimited space for the formation to move, in an experiment, the room's physical constraints must be taken into account. Therefore, the linear navigation of the simulation was changed to a periodic wave in one direction, meaning the system will move back and forth in a linear direction. Thus, the desired trajectory for the load is described by

$$\mathbf{x}_{0d} = \begin{bmatrix} \frac{3}{4}(1 - \cos \omega t) \\ 0 \\ 0 \end{bmatrix} + \mathbf{x}_0^{st}. \quad (84)$$

where, once again, $\mathbf{x}_0^t$ is the current position of the load after the safe take-off. The period, however, is changed to $T = 10$ seconds.

Furthermore, unlike the simulation, where the formation began with three agents and finished navigation with two, in the experiment, the formation starts with three agents, changes to two, and then returns to the original configuration. This was done to illustrate that the original formation can be recovered once the need for changing the number of agents is over. In the experiment, the first phase of the transition, where the robots assume the position depicted in Figure 3, was triggered manually via a joystick, while the second phase, where the third agent moves to the center of the platform, occurred 5 seconds later. The transition times were around $t = 6$ s and $t = 42$ s.

The complete experiment is shown in the video available at <https://youtu.be/qWt172HGMsc>. There, it is possible to see the different phases of the experiment. First, there is a safe take-off to ensure a stable condition, where all the cables are taut. After, the agents assume the desired relative poses to the load before beginning to track the trajectory. Then, the first transition begins, with the camera filming from behind, showing the system's behavior and the three robots lined up in a narrow formation. Afterward, the camera moves to a lateral view to show that the free agent is safely positioned between the other two agents, clear from their cables. Finally, from an oblique view, a new transition occurs, whereupon the agents assume their original relative position, and after a few seconds, the drones and the load land successfully. As one can notice, throughout all these transitions, the load continues to track the trajectory, traveling back and forth.

A snapshot of the navigation when $t = 22$ s is shown in Figure 21. This particular time instant was chosen because it is a moment when the formation is altered, changing to only two agents. It is possible to observe, when viewing from behind and in the video, that the formation is indeed much narrower. Imagining a corridor with walls at $y = 0$ m and $y = 1$ m, the whole formation could navigate through it. However, before this change, agents 2 and 3 would crash. This indicates that momentarily relinquishing control of the formation in the y -axis can be advantageous, as the movement is primarily in the x direction, which does not cause significant oscillation in the y direction. The side view shows once more that the free agent has positioned itself correctly between the two others.

The temporal evolution of the load errors is presented in Figure 22. Similar to the simulated results, solid rectangles indicate the transition period. Once again, the initial errors in the x -axis direction are larger, since that is the direction of maximum change in the trajectory. The error is around 15 centimeters in the x -axis, while in the other directions it depends on the navigation phase. In this experiment, it is more difficult to identify the steady error on the y -axis in the simulation, as the larger disturbances obscure it. However, the error seems to hover below the origin more than above.

The desired and the actual tensions are shown in Fig. 23, where it can be seen that they have similar behavior as the tension forces in the previous experiment, shown in Fig. 20. Once again, there is little difference between the desired and actual values. Unlike the previous experiment, however, the present one has two transition phases, during which the desired tension exhibits a discontinuity for all agents. Nonetheless, the figure shows that the system could adjust and follow these new desired values. Finally, for agent 3, the desired and actual tensions go to zero between the two transition periods. This happens because the cable attached to it is no longer taut, so it has no impact on the actual tension on the load.

Finally, this experiment demonstrates that the controllers presented in this work can be used to modify the formation's configuration to adapt to physical constraints with minimal impact on the load's navigation.

C. EXPERIMENT #3: CIRCULAR TRAJECTORY IN THE PRESENCE OF EXTERNAL DISTURBANCES

As a last experiment, it was decided to test if the system was resistant to external disturbances. To do so, the same conditions as the first experiment were repeated, with the difference being that a pole would be used to physically "push" elements of the formation, either the load itself or one of the drones. This would prevent the formation from achieving the desired trajectory, and once the disturbance was removed, the controller would have to rectify the errors.

The complete experiment can be watched at <https://youtu.be/bfZiwmLaWTU> for a better understanding. It shows the load following a circular trajectory while a researcher repeatedly uses a pole to disturb different elements. Visually, the system is somewhat malleable to disturbances but quickly returns to its correct position once the physical restriction is lifted. Figure 24 shows a depiction of the complete navigation, where the disturbances are visually present for all the agents of the formation.

The error evolution in time is present in Fig. 25. There, it can be seen that, even though the disturbances are very large in the video, they do not significantly impact the errors in the x - and y -directions. Their behavior is pretty similar to the one shown in Figure 19, except that this experiment is longer. The errors in z , however, indicate that the disturbance had a significant impact on this axis. Nonetheless, once the disturbance was removed, the controller quickly got the error under control, and it hovered near zero, with less than 10 centimeters of error.

The actual and desired tensions are shown in Fig. 26, where, as in the load errors graph, it shows similar behavior to the tensions of the first experiment. Once again, the first agent exhibits a more oscillatory response, likely due to its greater activity. Furthermore, while it can be seen that there were some disturbances, they are not as apparent as the ones in Fig. 25, showing the capacity of the controller to resist some level of real-world disturbances.

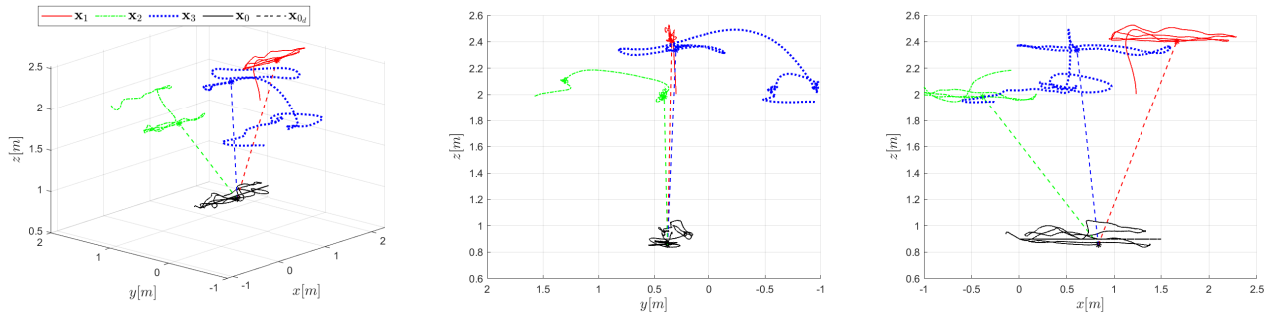


FIGURE 21. Navigation of the system in Experiment #2 at $t = 22s$, shown in three different views: aerial, rear, and lateral.

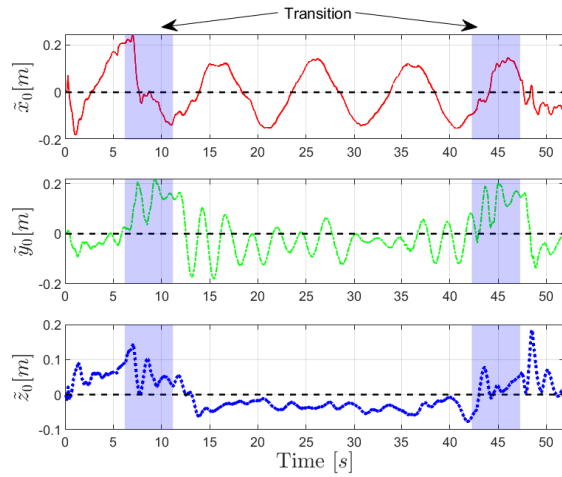


FIGURE 22. Performance of the load errors in Experiment #2.

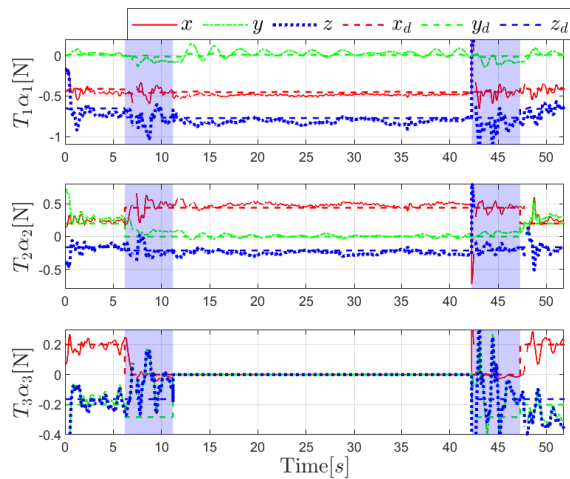


FIGURE 23. Performance of the actual and desired tension forces in Experiment #2.

Finally, as the experiment video and Figures 24-26 show, the formation was able to be malleable enough to not diverge once it was affected by external disturbances, and to return to its desired trajectory once these disturbances were gone.

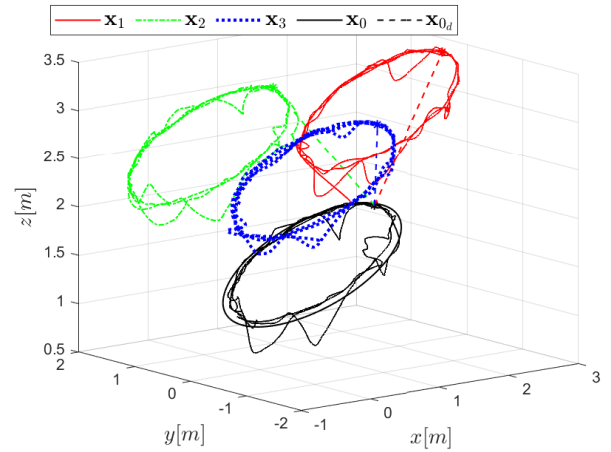


FIGURE 24. Navigation performance of Experiment #3.

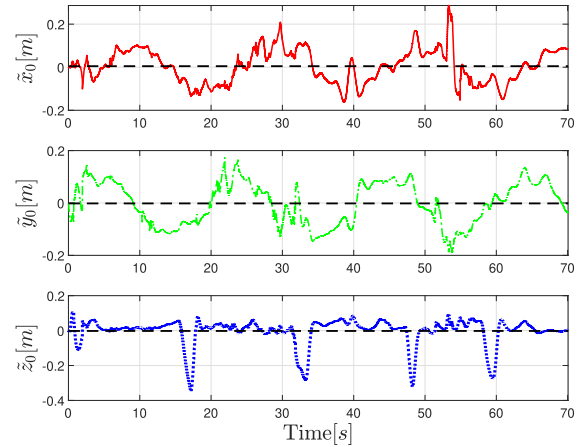


FIGURE 25. Performance of the load errors in Experiment #3.

It also presented minimal performance loss, as can be attested by comparing this experiment to the one in section VI-A.

D. RESULTS OVERVIEW

This section aims to summarize the main results and insights across the several results presented in the work. Overall, it is possible to claim that the system was capable of

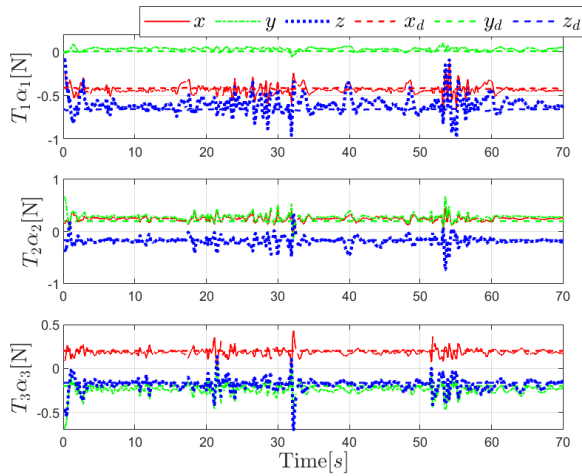


FIGURE 26. Performance of the actual and desired tension forces in Experiment #3.

fully accomplishing the tasks of tracking a trajectory and dynamically reconfiguring the number of agents during a mission.

The simulations conducted in this work demonstrated the effectiveness and scalability of the proposed control approach for cable-suspended load transportation by multiple UAVs. In the first simulation, a circular trajectory-tracking task with three UAVs demonstrated that the load successfully converged to its desired path, and the tension forces in the cables also converged to their desired values, validating the load tension control strategy. The second simulation demonstrated the leader-follower control strategy, where a third agent was repositioned to reduce the width of the formation. While initial errors increased during the transition, they were quickly controlled, indicating the system's ability to adapt to changes in formation configuration without severely impacting load navigation. Finally, the third simulation further validated the scalability of the proposed controller by demonstrating the successful tracking of a circular trajectory with five UAVs.

Meanwhile, the real-world experiments largely corroborated the promising simulation results, demonstrating the practical applicability of the proposed controllers. The first experiment, which involved a circular trajectory tracking with three UAVs, showed that the load followed the desired path, albeit with larger periodic errors compared to the simulation. However, despite these larger errors, the overall behavior of the system, including tension distribution, was consistent with the simulations. The second experiment replicated the dynamic reconfiguration of agents, transitioning from three to two and then back to three UAVs during a periodic linear trajectory. This experiment successfully illustrated the ability to change the formation's configuration to navigate physical constraints (like a narrow corridor) and then restore the original formation without significantly hindering the load's navigation. Meanwhile, the third experiment assessed the system's resistance to external disturbances by introducing

physical pushes to the load and drones. The results showed that the system exhibited malleability to the disturbances and quickly recovered its desired trajectory and attitude once the disturbance was removed. This experiment validated the controller's ability to resist real-world disturbances to a certain extent.

A primary advantage of our work, compared to others in the literature, is the ability to dynamically reconfigure the number of agents transporting the load during a mission, giving the system flexibility to adapt to physical constraints or agent failure. In addition, the controller explicitly closes the control loop on the load, rather than treating it as a disturbance. Another advantage is that the simulation shows a trajectory tracking task being accomplished with $n > 4$ agents, demonstrating the system's scalability.

As drawbacks, we identify the lack of robust or adaptive terms in the controller, which hinders applications in the real world, as the system has limited capacity to react to changes in its parameters and the environment. However, as the third experiment shows, the system has some margin to resist external disturbances. Another drawback, closely related to the previous one, is the lack of experiments conducted outdoors, which would help demonstrate how the system behaves in more realistic conditions, as well as without the use of a motion capture system. We plan to mitigate these drawbacks in future work.

VII. CONCLUSION

The paper proposed a controller for trajectory tracking of a load carried by multiple drones via cables. This was achieved by utilizing the resultant tension in the load as a virtual input, which was then transmitted to each aerial vehicle according to the desired geometric configuration. This, in turn, was transformed into a desired position and attitude to each drone, which was accomplished through low-level local controllers. Furthermore, a leader-follower-based controller was employed to safely reconfigure the UAV formation carrying the load, ensuring it conformed to physical constraints. It dealt with the load as the leader, and each free agent was a separate follower. The agents would position themselves in a desired offset from the leader while following the whole formation.

The simulations yielded promising results, with the system behaving as predicted and successfully accomplishing the missions. Furthermore, the system has also behaved as expected in the experiments. In the first experiment, the load tracking error was approximately 20 centimeters for the x - and y -axes, and around 5 centimeters for the z -axis. The tension values also behaved similarly to those presented in the simulation. The second experiment, on the other hand, demonstrated the successful application of reconfiguring the formation by changing the number of agents during the mission. It also showed that the reconfiguration of the formation did not hinder the navigation of the load, with the errors having a similar magnitude as the previous experiment. Lastly, the third experiment demonstrated that the system can

tolerate a certain degree of disturbance without significantly compromising its performance.

Finally, as part of future work, we plan to enhance the load controller to address its current limitations, particularly in terms of adaptability and robustness. For instance, the mass of the load is currently assumed to be known and constant throughout the mission. Any variation in this parameter may compromise trajectory tracking performance. To mitigate this issue, an adaptive control component could be introduced to adjust the controller online in response to changes in system parameters.

In addition, robust control strategies, such as sliding mode control, are being considered to improve the system's resilience to external disturbances, including wind, which are especially relevant in outdoor environments. Alongside these strategies, new sensing solutions are being investigated, including cameras, Ultra Wideband (UWB) sensors, and RTK GPS, with the aim of enabling reliable operation in outdoor scenarios and bringing the system closer to real-world deployment. Finally, strategies to prevent intra-formation collisions are also being explored, including the use of repulsive virtual forces designed to inhibit the close approach of any two agents.

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VINÍCIUS PACHECO BACHETI received the B.S. degree in electrical engineering from the Federal University of Viçosa, Brazil, in 2016, and the M.Sc. degree in electrical engineering from the Federal University of Espírito Santo, Brazil, in 2021, where he is currently pursuing the Ph.D. degree in electrical engineering. His research interest includes the control of multi-robot systems applied to load transportation with aerial robots.



DANIEL K. D. VILLA received the degree in electrical engineering from the Federal University of Viçosa (UFV), the master's degree in agricultural engineering from UFV, focusing on synthesizing and implementing controllers for precision agriculture, in 2017, and the Ph.D. degree in electrical engineering from the Federal University of Espírito Santo (UFES), in 2022, specializing in adaptive and sliding mode control applied to aerial robots, particularly cargo transportation.

He conducted research in control and automation with UFV. He is a Professor with the Department of Electrical Engineering, UFES. He is an expert in robotics, control, and automation, with research interests in control of multi-robot systems, nonlinear control, optimal control, and state estimation.



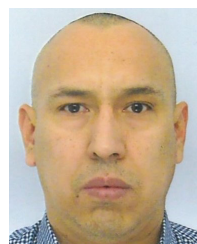
ROGELIO LOZANO (Senior Member, IEEE) was born in Monterrey, Mexico, in 1954. He received the B.S. degree in electronic engineering from the National Polytechnic Institute, Mexico City, Mexico, in 1975, the M.S. degree in electrical engineering from the Centro de Investigación y de Estudios Avanzados (CINVESTAV), Mexico City, in 1977, and the Ph.D. degree in automatic control from the Laboratoire d'Automatique de Grenoble, Grenoble, France, in 1981. From 1981 to 1989,

he was with the Department of Electrical Engineering, CINVESTAV, where he was the Head of the Section of Automatic Control, from 1985 to 1987. He held visiting positions with The University of Newcastle, Australia, from 1983 to 1984; the NASA Langley Research Center, Langley, VA, USA, from 1987 to 1988; and the Laboratoire d'Automatique de Grenoble, from 1989 to 1990. He was promoted to first-class CNRS Research Director, in 1997. From 1995 to 2007, he was the Head of the Laboratory Heudiasyc, CNRS UMR 6599, University of Technology of Compiègne (UTC), Compiègne, France. Since 1990, he has been the CNRS Research Director with UTC. He is currently the Head of the Laboratorio Franco-Mexicano de Informática y Automática Aplicada, UMI CNRS, CINVESTAV. His current research interests include adaptive control of linear and nonlinear systems, robot manipulators, passive systems, teleoperation, and uncrewed aerial vehicles. From 1987 to 2000, he was an Associate Editor of *Automatica*. He has been an Associate Editor of *International Journal of Adaptive Control and Signal Processing*, since 1993.



MÁRIO SARCINELLI-FILHO received the B.S. degree in electrical engineering from the Federal University of Espírito Santo, Brazil, in 1979, and the M.Sc. and Ph.D. degrees in electrical engineering from the Federal University of Rio de Janeiro, Brazil, in 1983 and 1990, respectively. He has been with the Department of Electrical Engineering, Federal University of Espírito Santo, since 1979, where he is currently a Professor. He has authored two books and 17 book chapters

and co-authored over 75 peer-reviewed journal articles and over 380 peer-reviewed conference papers. Considering the citations of his papers, his H-index is 26 in the Web of Science database, 31 in the SCOPUS database, and 38 in the Google Scholar database. He has also advised 22 Ph.D. and 29 M.Sc. students. His research interests include nonlinear control, mobile robot navigation, coordinated control of mobile robots, uncrewed aerial vehicles, multi-articulated robotic vehicles, and coordinated control of ground and aerial robots. He is also a Researcher with Brazilian National Council for Scientific and Technological Development (CNPq); a member of the Executive Advisory Board of *Journal of Intelligent and Robotic Systems* (Springer); an Associate Editor of *Journal of Control, Automation and Electrical Systems* (Springer); and a member of the International Advisory Committee of the International Conference on Uncrewed Aircraft Systems (ICUAS).



PEDRO CASTILLO (Member, IEEE) received the B.Sc. degree in electromechanic engineering from the Instituto Tecnológico de Zacatepec, Zacatepec, Mexico, in 1997, the M.Sc. degree in electrical engineering from the Centro de Investigación y de Estudios Avanzados (CINVESTAV), Mexico City, Mexico, in 2000, and the Ph.D. degree in automatic control from the Université de Technologie de Compiègne (UTC), Compiègne, France, in 2004. He has held several visiting positions

with The University of Sydney, Sydney, NSW, Australia; Massachusetts Institute of Technology (MIT), Cambridge, MA, USA; the Polytechnic University of Valencia, Valencia, Spain; Virginia Tech, Blacksburg, VA, USA; and LAFMIA UMI CNRS, Mexico City. He is currently the Research Director with the Heudiasyc Laboratory, French National Research Center (CNRS), UTC. His research topics include nonlinear control, delay systems, predictors, observers, autonomous vehicles, data fusion, robust navigation, real-time control, and robotics.

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